Role of serotonin transporter function on alcohol use in adolescent rats and their offspring

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Abstract

The study looked at serotonin dysfunction on ethanol consumption in adolescents and subsequent transgenerational influences of drinking using animal-models: SERT$^{+/+}$ (regular functioning serotonin transporter), SERT$^{+/-}$ (50% reduction) and SERT$^{-/-}$ (complete reduction). Adolescent rats showed greater development of ethanol seeking behaviours and no differences in consumption based on genotype were found in males. However, female rats showed patterns of drinking akin to those found in humans with reduced 5-HT activity. This was seen in the SERT$^{-/-}$ having the highest consumption and suggests a difference in the way the same genetic factors act in different genders. Examination of offspring of high ethanol drinking rats revealed a paternal genotype x offspring genotype interaction. SERT$^{-/-}$ offspring of SERT$^{-/-}$ fathers showed decreased sensitivity to the behavioural effects of ethanol indicating protective transgenerational changes. Thus highlighting the need to look at epigenetic and transgenerational effects of alcohol.

Role of serotonin transporter function on alcohol use in adolescent rats and their offspring

Alcohol is the most commonly consumed addictive substance. The Ministry of Health in their 2016 report claimed 80% of the New Zealand general population (aged above 15 years) had used alcohol, at least once, in the past year. Additionally, 19% of New Zealanders showed hazardous alcohol consumption with this being highest among 18-25 year olds (Ministry of Health, 2016). Hazardous drinking is characterized by a high frequency, quantity and irregular timing of drinking (Babor, Higgins-Biddle, Saunders, & Monteiro, 2001). Alcohol is an associated risk factor for over 60 different disorders with the socio-economic repercussions, making it the most harmful drug of abuse (Van Amsterdam et al., 2013). The need to investigate the nature of alcohol consumption is highlighted by an estimated 50% increase in alcohol related deaths between 1990 and 2010 (Lozano et al., 2012).

Despite large numbers of people showing hazardous consumption, compulsive alcohol use directly harms about 9.6% of users in younger populations, aged 16 to 25 years (National Committee of Addiction Treatment, 2011). The issue of hazardous drinking in adolescence is considered a risk factor for the development of alcohol dependence (AD) and other psychiatric disorders in adulthood (Witt, 2010). The aim is understanding why only certain people transition from normal, recreational alcohol use to AD? Understanding the neural and physiological mechanisms that determine the development of alcoholism is a crucial step towards the development of innovative prevention and treatment strategies for a problem which bears high prevalence and costs to society.

In regards to development of compulsive drug seeking behaviours, the neurotransmitters present in the brain's reward system are of the most interest. Among them

Serotonin, or 5-hydroxytyptamine (5-HT), plays an essential part in complex novelty-seeking behaviours and also in the negative effects of withdrawal both of which are associated with development and maintenance of drug use (Hayes & Greenshaw, 2011). 5-HT is produced in the raphe nuclei, its release is influenced by the value of expected and received rewards and has a modulating function on all major neurotransmitters including the reward-related dopamine (Hayes & Greenshaw, 2011). The serotonin transporter (SERT), on the presynaptic neuron, controls the reuptake of 5-HT released and thus regulates the duration of synaptic action. Alterations in these mechanisms could be responsible for several behavioural and neurocognitive dysfunctions. Early onset of AD among men is associated with a low serotonin turnover rate (Virkkunen & Linnolia, 1997). Pharmacological blockade of SERT with serotonin reuptake inhibitors (SSRIs) has demonstrated efficacy in the treatment of alcoholism for at least some people (Johnson.et al, 2004). The differential effects of SERT antagonism on alcohol consumption behaviours suggest variations in SERT function may play a role in influencing AD.

Genetic Links of AD

Family, twin and adoption studies have shown that drug dependence and alcohol addiction bear genetic links (Kendler, Jacobson, Prescott & Neale, 2003). It has been suggested that these compulsive behaviours develop as a result of an enhanced impulsivity or an enhanced sensitivity to the negative effects of drug withdrawal (Everitt, 2014).

Serotonergic dysfunction is often cited as a risk factor in the development and maintenance of AD (Heinz, Higley, Gorey & Saunders, 1998). Genetic influencers for SERT have been extensively researched and variations in the gene for the serotonin transporter protein (*SLC64A*) have been associated with ethanol sensitivity abnormalities and voluntary alcohol consumption (Thompson & Kenna., 2016). The *SLC64A* moderates the reuptake of

5-HT regulating synaptic levels of the neurotransmitter. Traditionally there are two variants of the 5- HTTLPR (5-hydroxytryptamine linked polymorphic region); the 16-repeat long (L) variant; and the 14-repeat short (S) variant (Lesch, 1996). The L-allele provides higher serotonin transporter mRNA transcription resulting in increased density and functional capacity when compared to the S-allele (Lesch, 1996). The S-allele causes a 50% reduction in SERT activity which results in decreased 5-HT turnover, caused by the decreased re-uptake of 5-HT which has been associated with ethanol abuse (Heinz et al., 1998). A meta-analysis of seventeen studies found that patients diagnosed with AD were 18% more likely to carry the S-allele than controls and this increased to 34% for patients with additional complications like early onset AD or a co-morbid psychiatric conditions (Feinn, Nellissery, & Kranzler, 2005).

Moreover, the S-allele is associated with disorders comorbid with drug addiction. The allele is strongly associated with neuroticism, which is the tendency to experience negative emotional states like anxiety and depression, which are disorders often comorbid to AD (Russo & Nestler, 2013). Another meta-analysis of 38 studies looking at ethanol abuse found an association between the S-allele and AD in combined European and Asian populations (Cao, Hudziak & Li, 2013). This association was with increased ethanol intake over time, rather than with increased initial intake suggesting chronic use of alcohol is what puts people with the S-allele at risk of AD (van der Zwaluw et al., 2010).

However, the role of 5-HTTLPR in AD is still debatable as Villalba et al. (2015) in a meta-analysis using twenty-five matched case-control studies found no association between the 5-HTTLPR and AD, while, AD was associated with the L-allele in an African population (Cao et al. 2013). Korean AD inpatients also had a significantly higher frequency of the L-allele compared to controls (Kweon et al., 2005). Suggesting that under certain conditions the allele may be protective against AD development (Krishnan et al., 2015).

These inconsistencies may be partially linked to variations between sample populations (ethnic differences) and methodologies (differences in patient diagnosis), thus not allowing direct comparisons. In addition, there is now evidence for *at least* a triallelic nature of 5-HTTLPR (Enoch et al., 2011), this is due to an additional single nucleotide polymorphism within the L-allele. The L-allele has been subtyped into high and low SERT activity, such that the pairing activities differ to what were initially perceived; S/S, S/LG, LGLG = low activity; S/LA, LA/LG = medium activity; and LA/LA = high activity. Many previous studies fail at making this distinction which may cause confounds (Cao, et al., 2013). While most studies have found an association between AD and the 5-HTTLPR, there is considerable heterogeneity within the clinical AD phenotype suggesting it has a complex aetiology; whereby the risk factors may vary with AD subtypes, comorbidities, age of onset, ethnicity and wider environmental factors.

There is evidence that the influence of the SERT on alcohol intake may be different between males and females. The 5-HTTLPR genotype was associated with alcohol consumption in the general population, but its effect depended on sex and birth cohort. It was found that the S/S female subjects had highest consumption particularly in the younger cohorts (Vaht et al., 2014). A significantly higher frequency of the L/L SERT genotype was found in female addicts without a known co-morbid psychiatric disorder than in the controls, which is opposite to the pattern found in most studies on males (Gokturk et al., 2008). This highlights the need of studies involving both sexes when looking at genetic links of disorders such as AD.

Transgenerational effects of alcohol

Adding to the complexity of the familial pattern of alcohol abuse are studies suggesting that epigenetic alterations may also be involved. Epigenetic changes refers to changes in gene expression due to for instance DNA methylation or histone modifications. In humans, several groups have now shown that children of fathers with AD have higher risk for psychosocial abnormalities, including increased risk for psychiatric disorders (Ozkaragoz, Satz, & Noble, 1997). Transmittable genetic alterations have been implicated in behavioural changes like enhancing the risk for alcoholism by increasing the tendency to consume alcohol and altering neurophysiological response to alcohol challenge (Nizhnikov, Popoola & Cameron, 2016).

Exposure to ethanol in either parent in rats have shown epigenetic effects. Nizhnikov et al. (2016) demonstrated that moderate ethanol exposure during late gestation (GD 17 to 20) in dams increased ethanol intake across at least 3 generations. Recent rodent studies have specifically focused on paternal transmission of epigenetic variants, which is especially relevant because sires are not present for offspring rearing and changes to offspring phenotype are assumed to result from modifications to the sperm epigenome (Finegersh, Rompala, Martin & Homanics, 2015). However, few studies have investigated the existence of an interaction between genetic and epigenetic factors.

The current study investigated whether parental alcohol consumption affected ethanol sensitivity in offspring, and whether this is influenced by the genotype of the parent and/or the offspring.

Animal Models

Genetic studies in humans are correlative in nature, and do not indicate clear causal relationships between the S-allele and the development of AD. Van der Zwaluw et al. (2010)

argue that the weak effect of genotype on ethanol intake in humans is possibly due to multiple confounding factors, like socioeconomic status and major life events (like parental behaviour itself). To allow for clear genetic causalities and control for confounders animal models allow easier interpretation.

Supporting the discussion on the role of serotonin in addiction, abnormally low levels of extracellular 5-HT were shown to be associated with increased preference for alcohol in non-human primates (Heinz et al., 1998) and rodents (Murphy et al., 2002). Additionally, a SERT mouse model showed increased inhibition of 5-HT clearance from the extracellular space after ethanol administration in mice with a reduction of SERT (Daws et al., 2006). The study found that genetic inactivation of SERT in a knock-out mouse not only failed to prevent ethanol-induced inhibition of 5-HT clearance, but actually potentiated this effect as seen through behavioural manifestations. However, there are fundamental differences between SERT models in rats and mice. For instance $SERT^{-/-}$ mice are significantly less sensitive to MDMA (3,4-methylenedioxy-N-methylamphetamine, ‘ecstasy’; Trigo et al., 2007), in sharp contrast to rats and humans (Oakly, Brox, Schenk & Ellenbroek, 2014)

Multiple rat models of alcohol consumption have been developed. These models of drinking include the ALKO alcohol/non-alcohol (AA/ANA) lines (Eriksson, 1968), the University of Chile A and B lines (Mardones and Segovia-Riquelme, 1983), the alcohol-preferring/non-preferring (P/NP) lines (Lumeng et al., 1977), and high/low alcohol-drinking (HAD/LAD) replicate lines (Li et al., 1993). All these were developed by selective breeding based on ethanol preferences. Murphy et al. (2002) looked at different rat models of alcohol drinking and found low levels of extracellular 5-HT corresponding with ethanol preference in some rats (HAD and P rats) but not in others. In fact AA rats showed increased preference with high levels of serotonin in extracellular regions and cortical structures. These results indicate that selective breeding for ethanol preference can yield a variety of phenotypes for high ethanol consumption with multiple underlying pathologies.

This study uses a SERT model to investigate escalation of alcohol intake in rats without preference-based breeding. SERT knock-out rats ($SERT^{-/-}$) were generated by N-ethyl-N-nitrosourea (ENU)-driven target-selected mutagenesis (Smits et al., 2006). The extracellular levels of 5-HT found in $SERT^{-/-}$ are 9 times greater than wild-type ($SERT^{+/+}$) littermates. This allows us to investigate a gene-dose response relationship. The heterozygous SERT knock-out rats ($SERT^{+/-}$) show, similar features to the homozygous S-allele in humans, which causes a 50% reduction in 5-HT Transporter ability (Homberg et al., 2007) making $SERT^{+/-}$ a valuable subject in our study. In line with human studies, rats with a genetically compromised SERT function show significantly more anxiety and depression-like symptoms, and are more sensitive to the rewarding properties of other drugs of abuse (Olivier et al., 2008).

Few studies have actually investigated ethanol consumption in both sexes. One study using $SERT^{-/-}$ mice found a slight reduction in alcohol intake when compared to their $SERT^{+/-}$ and $SERT^{+/+}$ counterparts (Lamb & Daws, 2013). They, reasoned the difference was that knock-outs were more sensitive to the negative effects of ethanol. On the other hand, Barr et al. (2004) showed an increased alcohol preference in female primates with a genetic reduction in the SERT. However, neither of these studies looked at compulsive alcohol intake. Here, we will compare all three SERT genotypes in both male and female rats, allowing us to investigate this further.

Current Study

Previous investigations of compulsive alcohol intake at our lab has resulted in relatively low consumption of alcohol by adult rats. To address this, we will be looking at adolescents rats. The reasoning for this is the increased translational validity to humans, as stated earlier, highest alcohol consumption is seen in adolescents and young adults (anywhere from 15- 25 years; Ministry of Health, 2016). The initial use of alcohol in children and adolescents is determined primarily by environmental influences and functional polymorphic variants in the promoter region of the 5-HTTLPR gene have age-dependent effects in alcohol abuse (de Oliviera et al., 2016).

The first part of the current study will examine ethanol drinking by SERT rats in intermittent access two bottle choice paradigm (IA2BC; Simms et al., 2008) to assess compulsive alcohol intake and the rewarding effects of ethanol. The paradigm taps into the ethanol deprivation effect (ADE), which is defined as a temporary increase in voluntary drinking over baseline when ethanol is reinstated after a period of deprivation (Heyser, Schulteis & Koob, 1997). Initial escalation is thought to be driven by repeated ADEs and neuro-adaptive changes that lead to sustained increases in baseline consumption following escalation (Simms et al., 2008). Importantly, intermittent access induces a transition from moderate to escalated and compulsive intake, which has some translational validity to the human AD initiation pattern (Simms et al., 2008). Subsequent to IA2BC, we will compare the rats for sensitivity to quinine. Quinine has an unpleasant taste and will significantly reduce liquid intake unless compulsive drinking has developed, which would cause the animals to ignore the aversive stimuli.

In the second part of the study, we will look at any transgenerational effects of alcohol. Despite evidence from several groups indicating that paternal ethanol exposure alters offspring development, no studies have examined its intergenerational effects on behavioural sensitivity to ethanol (Finegersh et al., 2015). Taking the offspring of the highest male and female drinkers from part one of our study, we will look for potential differences in the effects of ethanol using two well know assays: (i) ethanol-induced locomotor activity and (ii) ethanol-induced motor-coordination, via accelerating rotarod.

Hypothesis

Part one uses adolescent rats to study ethanol consumption patterns based on sex and genotype. Male SERT$^{-/-}$ animals have shown a greater sensitivity to the negative effects of ethanol compared to their SERT$^{+/+}$ counterparts (Heinz et al., 1998; Daws et al., 2006). However, female studies in humans have associated the S/S HTTLPR allele with highest consumption rates (Vaht et al., 2014). Therefore, we hypothesize a sex-genotype interaction with regards to compulsive consumption behaviours in SERT rats.

Part two looks at transgenerational effects of ethanol consumption. Since ethanol drinking was stopped before mating, any transgenerational and epigenetic effects sare hypothesized to be seen in the offspring of male drinkers. This is because males continue to go through cycles of spermatogenesis allowing epigenetic alterations being incorporated into the next generation (Finegersh et al., 2015).

Methods

Animals

Two cohorts of SERT Wistar rats were used (Smits et al., 2006). The mutation is a premature stop codon in the exon coding for the second extracellular loop of the transporter which leads to the absence of an mRNA transcript and protein in SERT$^{-/-}$ and reduced expression in SERT$^{-/+}$ (Olivier, 2008).

The first cohort consisted of Fifty ethanol naïve Wistar rats (Males *n*=24, Females *n*=26) which were tested for the rewarding properties of alcohol. Animals in this cohort were aged 35-37 days on the first day of testing and males (SERT$^{+/+}$ *n* = 8, SERT$^{-/+}$ *n* = 11, SERT$^{-/-}$ *n* = 5) weighed in at 168-213gms while females (SERT$^{+/+}$ *n* = 9, SERT$^{-/+}$ *n* = 11, SERT$^{-/-}$ *n* = 6) weighed between 130-168gms.

A second cohort of 82 rats (males *n*=38, females *n*=44) were used for ethanol induced locomotor activity and motor coordination. Subjects were a result of breeding the three highest drinking SERT$^{+/+}$ and SERT$^{-/-}$ males and females with naïve SERT$^{-/+}$ animals (figure 1). The cohort aged 25-30 days during locomotor activity testing, males weighed between 30-112gms while females weighed between 28-105gms. The same cohort aged 35-40 days during motor coordination testing with males weighing 83-231gms and females 85-175gms.

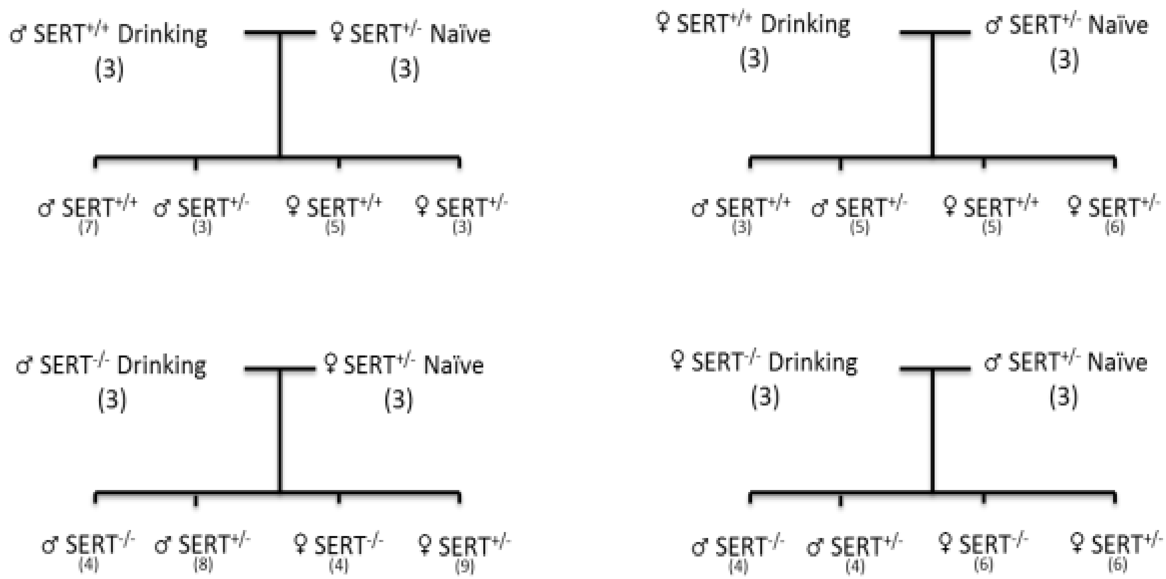

Figure 1. Cohort 2, used to assess transgenerational effects, rat lineage and genotypes. Numbers in brackets refer to number of animals of each genotype.

All subjects were bred at Victoria University of Wellington, New Zealand, vivarium. Animals were housed with same genotype littermates 3-4 per standard cage with access to food and water *ad libitum* in a temperature (20 ºC) and humidity (60%) controlled room where they were maintained on a 12-hour reverse light/dark cycle (lights off at 7am). During testing period cohort 1 animals were housed individually. After the experimental period animals were re-housed with same genotype litter mates 2-3 per standard cage. Cohort 2 were housed with same sex littermates regardless of genotype in groups of 2-7 per standard cage. Animal care procedures and experimental protocols met with Victoria University ethics committee approval.

Procedure

Intermittent-Access two bottle Choice Paradigm

Rats were trained to drink ethanol (20% v/v) in their home cages using an adapted version of the intermittent access two bottle choice paradigm (IA2BC) (Simms et al., 2008). Under this, rats were offered a bottle with ethanol for 7 hours/day for three days/week (Monday, Wednesday and Friday) for twelve sessions. On ethanol drinking days rats were offered two plastic bottles with stainless steel, drip proof, double ball bearing drinking spouts containing 20% (v/v) ethanol or tap water. Bottles were inserted on either side of the cage front and were presented 30 minutes after lights off and remained until 7 hours had passed. The ethanol bottle was replaced with a second water bottle until the next session began. After the initial four weeks, the rats were given access to ethanol for 24 hours/day for three days/week (Monday, Wednesday and Friday) for another 12 sessions. Ethanol was again presented 30 minutes after lights off and the bottle remained on for 24 hours after which it was replaced with water. Placement of ethanol bottles was alternated for each drinking session to control for side preferences. Ethanol and water intake were recorded by weighing the bottles after each session in order to calculate alcohol preference. Bodyweight was measured weekly to calculate g/kg intake. Following two and a half months of alcohol consumption, the three genotypes were compared for sensitivity to quinine adulteration

(Lesscher et al., 2010). Therefore, the ethanol solution was combined with six graded concentrations (0-1 gm/l) of quinine hydrochloride dihydrate in 24h drinking sessions during which volumes consumed were determined after 2, 7 and 24 hours.

Ethanol-Induced Locomotor Activity

Rats were introduced to the testing room 10-20 minutes before receiving an intraperitoneal (IP) injection of either saline, or 1.0 g/kg of 20% (v/v) ethanol solution. Doses were alternated within groups to control for tolerance and ordering effects. Animals were then placed in transparent Plexiglas open field (Med Associates, W x 43.2 x 43.2, H x 30cm) sound attenuated locomotor boxes with infrared beams. Activity was recorded for 60mins and processed by a computer programme (Activity Monitor Version 5 programme; Med Associates Inc., St Albans, VT USA). Horizontal locomotor activity was analysed as ambulatory counts measured by rats breaking the infrared beams across the arena. The testing room was dimly lit by a red light and a white-noise generator was used to disguise extraneous noise disturbance. On completion, rats were placed back in their home cages and returned to the home room. The apparatus was then cleaned with diluted Virkon antiseptic (1:100) (Antec International, Sudbury, UK). Testing was conducted between 12-5pm and rats were given a five day washout period between the two testing conditions.

Ethanol-Induced motor-coordination on the Accelerating Rotarod

Ethanol-induced motor-coordination was tested on an accelerating Rotarod treadmill (Rotarod LE 8500 (76-0239), L x 50, H x 36, W x 24cm, Panlan, Barcelona, Spain). The apparatus is a four station rotating roll (60mm diameter) mounted 10cm above four corresponding drop levers that register the time and speed in rotations per minute (rpm) at which an animal falls from the roll. The rat must balance on the roll and adjust to increasing speeds from 4 rpm to 40 rpm over a 5min period. For training days, animals were habituated to the testing room for 10-15min. Over the three training days each animal was given three trials daily with 5-10min inter-trial intervals. The aim of the training was to build up to 12 rpm. The last training day involved increasing the speed of the rotarod every 30 seconds with a minimum inclusion criteria of 15 seconds being set. Forty-eight hours after the last training session, each animal received an IP injection of saline, 1.0 g/kg or 2.0g/kg of 20% (v/v) ethanol solution. Doses were counter-balanced and animals were given two days before being administered a different dose. Rats, were placed on the rotarod after 10min to allow ethanol to take effect and again after 30 minutes had passed, latency to fall was recorded over two trials at each of these time points. There were no exclusion criteria for any of the test conditions. Ethanol-induced ataxia scores were analysed by comparing latency to fall at each condition between parent genotypes, animal genotypes and sex.

Solutions

Ethanol 20% (v/v) was made through dilution of 99.8% ethanol solution (Purescience Ltd, Porirua New Zealand) with distilled water. Quinine solutions were formed by adding quinine hydrochloride dihydrate (Sigma- Aldrich Co., St. Louis, Mo, USA) in the required amounts to the already made 20% ethanol solutions.

Data Analysis

Data was analysed using the computer software IBM Statistical Package for the Social Sciences (SPSS). A generalised linear mixed-model analysis with an autoregressive covariance structure was used for ethanol consumption. This model was chosen in order to accommodate for days during which data was lost due to leaks in the drinking bottles (Wang & Goonewardne, 2004). Independent variables used were three levels of genotype ($SERT^{+/+}$, $SERT^{+/-}$ and $SERT^{-/-}$) and two levels of sex (females and males) while ethanol consumed in grams per kilogram (g/kg) was the dependent variable. Quinine adulteration, locomotor activity and motor coordination were all measured using mixed factorial ANOVAs (analysis of variance), as no data points were missing. For locomotor activity and motor coordination, the drinking parent genotype was added as an additional independent variable. Parents with the same genotype were compared against each other (SERT$^{+/+}$ fathers vs mothers; SERT$^{-/-}$ fathers vs mothers). Dependent variables were ethanol mixed with quinine dose consumed in grams per kilogram, number of ambulatory counts recorded at each dose (locomotor activity) and latency to fall from the accelerating rotarod treadmill at each dose (motor coordination).

Results

Voluntary home cage ethanol consumption.

An intermittent access paradigm was used to investigate the rewarding properties of alcohol. Drinking behaviour was assessed over twelve 7 hour sessions followed by twelve 24 hour sessions. Drinking was significantly different between the two time durations (F (1, 44) = 55.18, p <.001, η_p^2 = .564) with an average increase in drinking from 7 hour sessions to 24 hour sessions (figure 2a & 2b). The general linear mixed models analysis revealed a significant main effect of sex (F (1, 147.174) = 15.19, p <.001, η_p^2 = .094) and also a significant sex x genotype interaction (F (2, 147.177) = 4.48, p =.013, η_p^2 =.057). There were no significant genotype main effects. Pairwise comparisons revealed female rats consistently consumed more ethanol per bodyweight compared to males (see Appendix A.). The interaction revealed SERT$^{-/-}$ females consumed the most ethanol followed by SERT$^{-/+}$ and SERT$^{+/+}$ females. In males, no significant differences were found between genotypes with SERT$^{-/-}$ males consuming similar amounts to SERT$^{-/+}$ and SERT$^{+/+}$ being the highest drinkers (figure 3a & 3b). These results suggest that while SERT$^{-/-}$ females drank more than SERT$^{+/+}$ females, no genotype difference in drinking behaviours was observed in males.

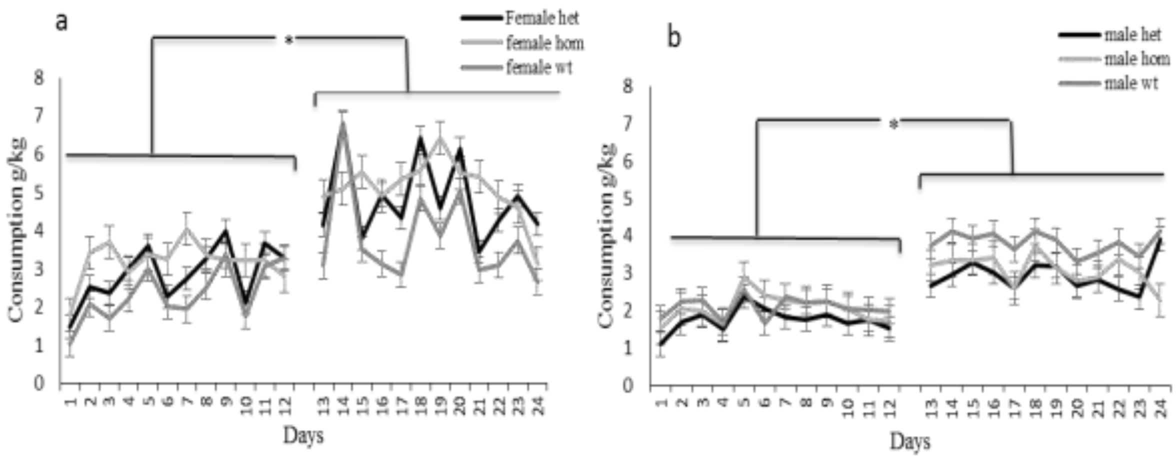

Figure 2. Ethanol consumption g/kg increased from 7 hour sessions (days 1-12) to 24 hour sessions (days 13-24) in both females (a) and males (b). Error bars denote standard error. *p<.05

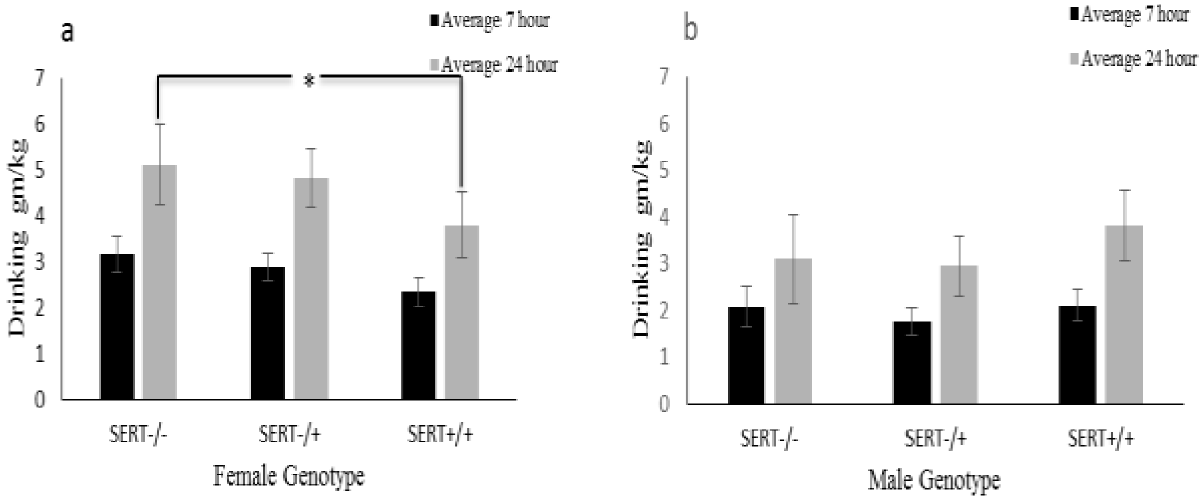

Figure 3. There was a sex genotype interaction where SERT$^{-/-}$ females showed the highest voluntary consumption followed by SERT$^{+/-}$ and SERT$^{+/+}$ (a) while no such pattern was observed in males. *p<.05

Compulsive ethanol seeking during quinine adulteration

Compulsive ethanol seeking was looked at over six doses of quinine being dissolved in increasing order over the same number of sessions. A 2 (sex) x 3 (genotype) x 6 (dose) mixed factorial ANOVA revealed a significant effect of dose (*F* (1, 40) = 71.68, *p* <.001, η_p^2 = .644) showing a reduction in consumption as the concentration of quinine increased (figure 4a & 4b). Sex differences were observed (*F* (1, 40) = 4.62, *p* =.038, η_p^2 = .103) with again females consuming higher quantities than males. However, there was no significant interaction between genotype and dose, indicating that the quinine induced decrease in consumption was similar in all genotypes.

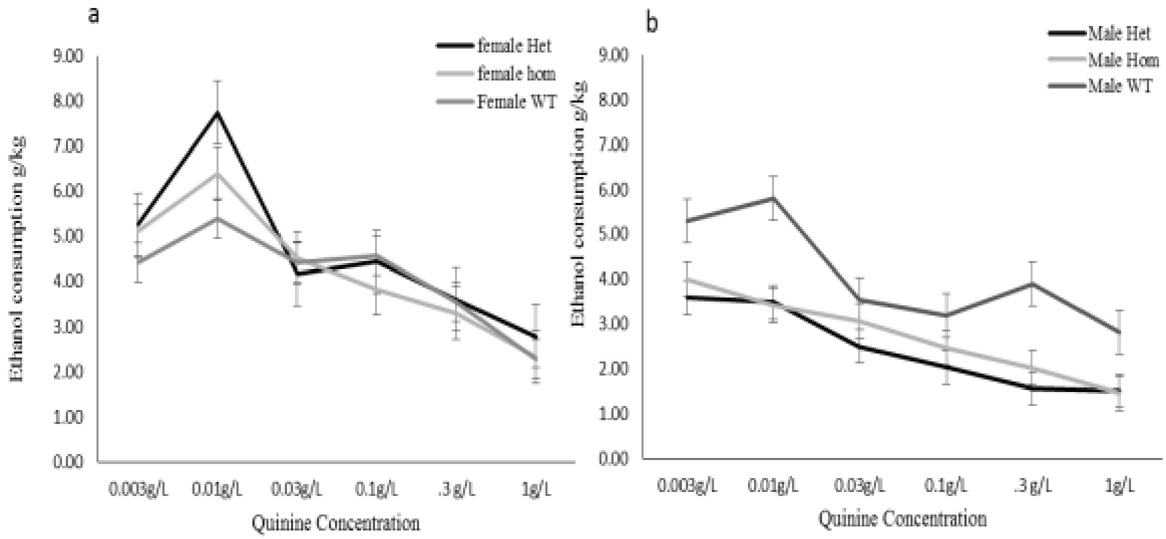

Figure 4. Ethanol consumption reduced with increase in dose. There was no statistically significant difference in compulsive ethanol intake between genotypes. Females (a) still consumed more per bodyweight than males (b).

Transgenerational effect of alcohol consumption

As described in the methods section, subsequent to the voluntary alcohol intake experiment, the 3 highest SERT$^{+/+}$ and SERT$^{-/-}$ male and female drinkers were mated with non-drinking naïve SERT$^{+/-}$ animals and their offspring tested for ethanol induced locomotor activity and motor coordination via rotarod.

Ethanol-Induced Locomotor Activity

Ethanol-induced locomotor activity was assessed in a novel environment following intraperitoneal injections of saline or 1.0 g/kg ethanol. A 3 (genotype) x 2 (sex) x 2 (dose) mixed factorial ANOVA revealed that there was no effect of genotype or dose. However, there was a significant sex main effect (F (1, 76) = 5.14, p =.026, η_p^2 =.063). Pairwise comparisons revealed, in the saline condition, female animals showed greater overall activity than males (Table 1). This remained consistent in the ethanol condition with males being less active compared to females.

Table 1.

Descriptive statistics for locomotor activity.

		Saline			1g/kg Ethanol		
Genotype	Sex	*M*	*SD*	*N*	*M*	*SD*	*N*
$SERT^{+/+}$	female	1173.40	234.73	10	1202.60	234.78	10
	male	914.50	140.07	10	1101.90	440.34	10
$SERT^{+/-}$	female	1282.21	299.52	19	1228.05	346.95	19
	male	1158.45	330.07	20	1182.90	444.72	20
$SERT^{-/-}$	female	1395.80	373.50	15	1233.55	289.51	15
	male	1218.50	249.05	8	1157.55	397.88	8

Note: M refers to mean ambulatory counts, *SD* refers to standard deviation and *N* refers to sample size of group.

Transgenerational effects of parents were analysed for drinking $SERT^{-/-}$ and $SERT^{+/+}$ fathers and mothers using a 2(parent) x 2(genotype) x 2(dose) mixed factorial ANOVA. In, $SERT^{-/-}$ parents, a parent main effect (F (1, 41) =8.27, p=.006, η_p^2 =.168) with offspring of drinking $SERT^{-/-}$ mothers being more active than that of drinking fathers (Figure 4).

Importantly, a significant dose x parent x genotype interaction was found (*F* (1, 41) =5.51, *p* =.024, η_p^2 =.118). Analysis of this showed that SERT$^{-/-}$ offspring of SERT$^{-/-}$ fathers showed no significant change in locomotor activity when injected with ethanol, while there was a decline in SERT$^{-/-}$ offspring of SERT$^{-/-}$ mothers, indicating that the effects of alcohol are diminished in offspring of fathers who have consumed (high levels of) alcohol. No significant main effect or interactions were found in the offspring of SERT$^{+/+}$ parents (see Appendix B).

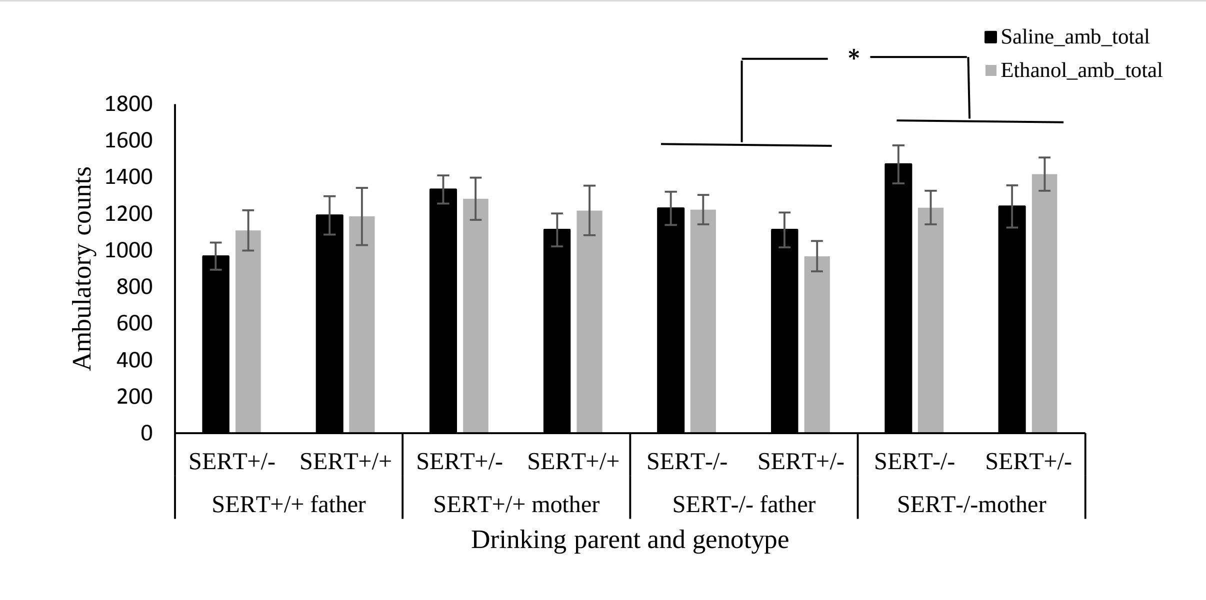

Figure 4. Locomotor activity by genotype and parent genotype. There was no statistically significant difference between the saline and 1g/kg ethanol dose. There was a significant difference between the ambulatory counts of SERT$^{-/-}$ mothers and SERT$^{-/-}$ fathers. *p=.024

Ethanol-Induced Motor coordination on the Accelerating Rotarod

Coordination was measured as the latency to fall from an accelerating rotarod treadmill. A 2(sex) x 3(genotype) x 3(dose) mixed model ANOVA revealed a significant main effect of dose (*F* (2, 152) = 22.92, *p* <.001, η_p^2 =.232), sex (*F* (1.76) = 3.86, *p* =.053, η_p^2 =.048) and genotype (*F* (2, 76) = 5.44, *p* =.006, η_p^2 =.125). Simple main effects analysis showed all animals had significantly decreased latency to fall after a 1.0g/kg IP injection of ethanol compared to saline (p <.001) and this declined further after a 2.0g/kg (p =.001). Pairwise comparisons of genotype revealed that SERT$^{-/-}$ animals were more active compared to SERT$^{+/-}$ (p =.002). SERT$^{+/+}$ animals had a lower average latency to fall to SERT$^{-/-}$ animals although this effect failed to reach statistical significance (p =.096).

Like locomotor activity, transgenerational effects of parent were analysed as offspring of drinking fathers and mothers of the same genotype using a 2(parent) x 2(genotype) x 3(dose) mixed factorial ANOVA. In SERT$^{-/-}$ parents, a dose x parent x genotype interaction was found (F (1.94, 79.65) = 3.94, p =.024, η_p^2 =.088). Between subjects effects showed a genotype main effect (F (1, 41) = 11.17, p =.002, η_p^2 =.214) and a parent x genotype interaction (F (1, 41) = 9.13, p=.004, η_p^2 =.182). There was no main effect of parent. SERT$^{-/-}$ offspring had a significantly longer latency to fall compared to SERT$^{+/-}$ (see Appendix C). Separate ANOVAs were used to understand the 3 way interaction. When analysing parent sexes for SERT$^{-/-}$ parents, no difference in motor coordination in offspring genotypes for SERT$^{-/-}$ mothers was found. There was no dose x genotype interaction in the offspring of SERT$^{-/-}$ mothers either. However, in SERT$^{-/-}$ fathers there was a significant difference in performance (F (1, 23) =26.25, p <.001, η_p^2 =.533) with SERT$^{-/-}$ offspring staying on the rotarod for longer (figure 5). There was a significant dose x genotype interaction (F (2, 46) =4.13, p =.022, η_p^2 =.152) in SERT$^{-/-}$ fathers with SERT$^{-/-}$ offspring showing a greater decline in performance compared to SERT$^{+/-}$ offspring. In SERT$^{+/+}$ parents, there were no effects of parent sex, genotype or any interactions (See Appendix D). Together, these data show that the offspring of drinking SERT$^{-/-}$ fathers are less sensitive to the effects of alcohol than the offspring of drinking SERT$^{-/-}$ mothers.

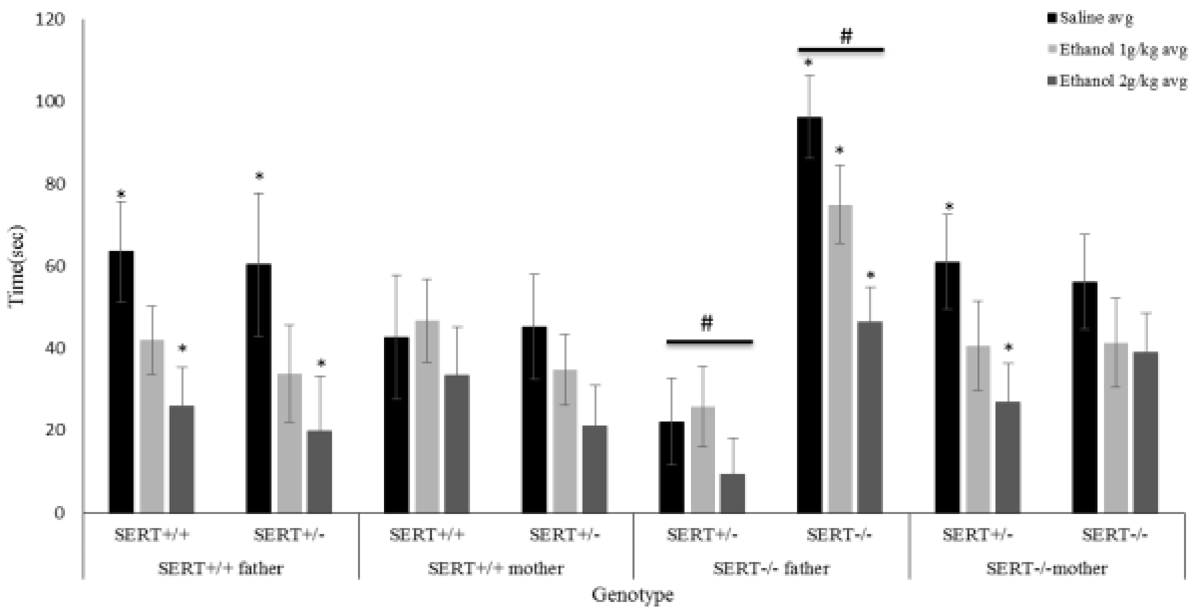

Figure 5. Motor coordination by latency to stay on rotarod in seconds organised by parent and offspring genotype. A reduction of coordination with increase of ethanol dose. * Significant difference in dose within genotype. # Significant difference of genotypes within parent.

Discussion

The two part study investigated whether the reduction of serotonin function induces differential ethanol consumption patterns in adolescent rats and, next, whether ethanol drinking in parents leads to transgenerational effects with regards to sensitivity to ethanol in the subsequent generation. First, we hypothesised a sex x genotype interaction as previous literature on ethanol drinking had shown $SERT^{-/-}$ male rats being more sensitive to ethanol (Heinz et al., 1998) while human females carrying the S/S HTTLPR allele showed highest rates of consumption (Vaht et al., 2014). Our findings support this hypothesis with $SERT^{-/-}$ females consuming the most ethanol followed by $SERT^{-/+}$ and $SERT^{+/+}$ females while no genotype difference in drinking behaviours was observed in males. Females also consumed more ethanol per bodyweight. Quinine reduction showed compulsive drinking behaviours did not depend on genotype and females again consumed higher quantities per bodyweight.

Second, with regards to transgenerational effects, we hypothesised epigenetic effects only in offspring of male drinkers based on males continuing cycles of spermatogenesis allowing epigenetic alterations (Finegersh et al., 2015) and female epigenetic influences being seen only during late gestation (Nizhnikov et al., 2016). Results partially supported our hypothesis as data showed that $SERT^{-/-}$ offspring of drinking $SERT^{-/-}$ fathers were less sensitive to the effects of alcohol than the offspring of drinking $SERT^{-/-}$ mothers. However, this difference was not seen in the offspring of $SERT^{+/+}$ parents.

Unlike previous attempts with adult rats in our lab, adolescents showed a pattern of escalation in consumption over time similar to Simms at al. (2008). Consumption increased when moving from the 7 hour to the 24 hour presentation sessions after which it remained stable. This aligns with research in humans showing interaction with alcohol at younger ages being a risk factor for development of AD (Van Amsterdam et al., 2013). Ethanol consumption on average was slightly lower compared to previous studies (Simms et al., 2008; Murphy et al., 2002). This could be attributed to differences seen between rat strains in ethanol consumption (Simms et al., 2008) and the current rats being adolescents while previous literature used adults. It would be interesting to analyse whether consumption increases further in animals, who have been exposed to ethanol in adolescence.

Genotype differences in drinking were seen only in females. Similar to female humans, those with reduced 5-HT activity showed the highest consumption rates (Vaht et al., 2014). Female rats consumed more ethanol per bodyweight but their day to day consumption showed greater fluctuations. Taken together, these suggest a clear difference in drinking patterns between sexes. This makes it important to look at both sexes when carrying out genetic studies as key differences could remain hidden.

One explanation for higher consumption may be due to patterns of isolation and depression. The individual housing of rats, may be a cause for the drinking patterns seen, especially with regards to depression particularly in females (Erol & Krapyak, 2015). Female adolescent AD is often comorbid with depression while this pattern is not seen in males (Danzo, Connell & Stormshak, 2017). Additionally, there are higher rates of adolescent AD leading to comorbidity with depression in adulthood in females (Erol & Krapyak, 2015). This isolation makes it harder to translate to humans as well because alcohol drinking is associated with social interaction (Lozano et al., 2012). A control would be to develop methods to monitor drinking behaviours while animals are housed with cage mates which translates better to drinking patterns in humans. Additionally, we did not look at adult controls but based the drinking differences on previous attempts conducted with the same rat strains.

With regards to the transgenerational studies, female offspring of drinking parents were more active than males regardless of dose in both locomotor activity and motor coordination which is consistent with previous studies looking at overall activity in rats (Fernandes, Gonzalez, Wilson & File, 1999). Genotype differences were found in motor coordination but not in locomotor activity. $SERT^{-/-}$ demonstrated the longest latency to fall off the rotarod in all dose conditions. However, they also showed the steepest drop in performances with increase of ethanol dose which is similar to the findings of Boyce-Rustay et al. (2006), who showed significant motor impairments in 5-HT KO mice compared to wild-types on the accelerating rotarod. This should be taken with caution as our $SERT^{-/-}$ rats, despite the greater relative drop, showed better performances. Interestingly, on locomotor activity there was no significant drop in performance between saline and 1g/kg ethanol dose. This could be attributed to studies showing no significant differences of locomotor activity in young rats (PND 15) when injected with moderate doses of ethanol (1.25 g/kg) (Arias, Mlewski, Molina & Spear, 2009) which is higher than what we administered here. The same study also showed an interaction with age as sensitivity increased with age, explaining why these differences became clearer by the time rats were tested on motor coordination.

Transgenerational analysis showed SERT$^{-/-}$ offspring of drinking SERT$^{-/-}$ fathers were less sensitive to the effects of alcohol than the offspring of drinking SERT$^{-/-}$ mothers. The same results were not seen in the SERT$^{+/+}$ parents indicating a specific relationship between genotype and protective transgenerational effects. These results are similar to male offspring of cocaine-exposed sires displaying a cocaine-resistance phenotype (Vassoler, White, Schmidt, Sadri-Vakili, & Pierce, 2013). Expanding this, recent studies in humans have found epigenetic links of AD and development of protective mechanisms in certain children of parents with substance abuse problems (Krishnan et al., 2015). Currently, our cohort 2 is going through the same IA2BC paradigm and we are interested to look at whether the reduced sensitivity to ethanol in the SERT$^{-/-}$ offspring of males will lead to increased consumption and whether ethanol sensitivity has increased in the SERT$^{+/+}$ offspring. This may indicate different directional outcomes in behaviour of offspring of ethanol exposed rats. The use of drinking female mothers as controls at the current time are backed up as females were not given ethanol from a month before pregnancy.

Immediate follow-ups involve looking at the biological aspects of the effects of ethanol consumption on progeny. Brains and testes of the highest drinking animals from cohort one and brains of offspring from each of the litters were collected for analysis. This gives us a complete understanding of the epigenetic changes which occur due to ethanol exposure. While in the longer run stress the importance of including both male and female animals.

Conclusion

Our study looked at ethanol consumption in adolescent SERT knock out rats to check for the effects of serotonin dysfunction in ethanol consumption. Adolescents showed greater development of ethanol seeking behaviours. Female rats showed patterns of drinking similar to humans with reduced 5-HT activity animals ($SERT^{-/-}$) showing the highest consumption rates. However, no clear patterns were observed in males, suggesting a differential influence of genetic factors in genders. Looking at ethanol induced transgenerational effects on offspring of high ethanol drinking rats we see paternal effects as offspring of $SERT^{-/-}$ fathers showed decreased sensitivity to the behavioural effects of ethanol. This indicates possible protective epigenetic changes being inherited and highlights the need to look at the underlying mechanisms involved in greater detail.

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Appendix A.

Table A 1.

Descriptive statistics for ethanol consumption by gender and genotype.

		7 hour/day			24 hour/day		
Genotype	Sex	*M*	*SE*	*N*	*M*	*SE*	*N*
$SERT^{+/+}$	female	2.34	0.32	9	3.80	0.71	9
	male	2.10	0.34	8	3.82	0.76	8
$SERT^{+/-}$	female	2.88	0.29	11	4.83	0.64	11
	male	1.74	0.29	11	2.94	0.64	11
$SERT^{-/-}$	female	3.17	0.39	6	5.11	0.87	6
	male	2.06	0.43	5	3.10	0.96	5

Note: M refers to mean g/kg ethanol consumed, *SE* refers to standard error and *N* refers to sample size of group.

Appendix B

Table B 1.

Descriptive statistics for locomotor activity parent.

		Saline			1g/kg Ethanol		
Parent	Genotype	*M*	*SE*	*N*	*M*	*SE*	*N*
$SERT^{+/+}$ Father	$SERT^{+/+}$	967.92	73.98	12	1108.75	110.73	12
	$SERT^{+/-}$	1190.50	104.62	6	1185.33	156.61	6
$SERT^{+/+}$ Mother	$SERT^{+/+}$	1158.00	90.60	8	1217.50	135.63	8
	$SERT^{+/-}$	1332.36	77.26	11	1281.91	115.66	11
$SERT^{-/-}$ Father	$SERT^{+/-}$	1111.58	94.95	12	967.92	83.30	12
	$SERT^{-/-}$	1229.38	91.22	13	1222.15	80.03	13
$SERT^{-/-}$ Mother	SERT+/-	1239.30	104.01	10	1416.30	91.25	10
	SERT-/-	1470.30	104.01	10	1233.90	91.25	10

Note: M refers to mean ambulatory counts, *SE* refers to standard error and *N* refers to sample size of group.

Appendix C

Table C 1.

Breakdown of performances For SERT$^{-/-}$ parents on each dose of accelerating rotarod based on gender and genotype.

Parent	Genotype	*N*	*Dose*	*M*	*SE*
SERT$^{-/-}$ Father	SERT$^{+/-}$ ♂	8	saline	24.34	14.24
			1 g/kg ethanol	26.13	11.51
			2 g/kg ethanol	8.44	11.10
	SERT$^{-/-}$ ♂	4	saline	81.56	20.13
			1 g/kg ethanol	60.00	16.28
			2 g/kg ethanol	46.50	15.70
	SERT$^{+/-}$ ♀	9	saline	17.44	20.13
			1 g/kg ethanol	25.19	16.28
			2 g/kg ethanol	11.56	15.70
	SERT$^{-/-}$ ♀	4	saline	102.78	13.42
			1 g/kg ethanol	81.67	10.85
			2 g/kg ethanol	46.39	10.46
SERT$^{-/-}$ Mother	SERT$^{+/-}$ ♂□	4	saline	68.44	20.13
			1 g/kg ethanol	44.63	16.28
			2 g/kg ethanol	8.00	15.70
	SERT$^{-/-}$ ♂□	4	saline	40.50	20.13
			1 g/kg ethanol	25.06	16.28
			2 g/kg ethanol	30.13	15.70
	SERT$^{+/-}$ ♀	6	saline	55.96	16.44
			1 g/kg ethanol	38.04	13.29
			2 g/kg ethanol	39.54	12.82
	SERT$^{-/-}$ ♀	6	saline	66.46	16.44
			1 g/kg ethanol	52.29	13.29
			2 g/kg ethanol	45.17	12.82

Note: M refers to mean latency to fall (seconds), *SE* refers to standard error and *N* refers to sample size of group.

Appendix D

Table D 1.

Breakdown of performances For SERT$^{+/+}$ parents on each dose of accelerating rotarod based on gender and genotype.

Parent	Genotype	*N*	*Dose*	*M*	*SE*
SERT$^{+/+}$ Father	SERT$^{+/+}$ ♂	7	saline	60.32	15.22
			1 g/kg ethanol	40.07	12.30
			2 g/kg ethanol	25.00	11.86
	SERT$^{+/-}$ ♂	3	saline	79.08	23.25
			1 g/kg ethanol	23.50	18.80
			2 g/kg ethanol	7.08	18.12
	SERT$^{+/+}$ ♀	5	saline	68.00	18.01
			1 g/kg ethanol	44.70	14.56
			2 g/kg ethanol	27.25	14.04
	SERT$^{+/-}$ ♀	3	saline	41.58	23.25
			1 g/kg ethanol	44.08	18.80
			2 g/kg ethanol	32.42	18.12
SERT$^{+/+}$ Mother	SERT$^{+/+}$ ♂	3	saline	23.58	23.25
			1 g/kg ethanol	31.50	18.80
			2 g/kg ethanol	8.17	18.12
	SERT$^{+/-}$ ♂	5	saline	44.00	18.01
			1 g/kg ethanol	35.95	14.56
			2 g/kg ethanol	23.85	14.04
	SERT$^{+/+}$ ♀	5	saline	54.10	18.01
			1 g/kg ethanol	55.80	14.56
			2 g/kg ethanol	48.65	14.04
	SERT$^{+/-}$♀	6	saline	46.17	16.44
			1 g/kg ethanol	33.83	13.29
			2 g/kg ethanol	18.67	12.82

Note: M refers to mean latency to fall (seconds), *SE* refers to standard error and *N* refers to sample size of group.